

RAJAT SAVDEKAR

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EDUCATION

University of Southern California, Los Angeles

Master of Science, Computer Science

Coursework: Multimedia System Design, Database Systems, Machine Learning, Game Design

Aug 2024 – May 2026

GPA: 3.63

University of Mumbai, Mumbai

Bachelor of Engineering, Computer Engineering

Aug 2019 – May 2023

GPA: 3.7

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Backend & APIs: Node.js, Flask, FastAPI, REST APIs, WebSockets

AI: LLMs, RAG, OpenAI Whisper, prompt engineering

Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, event sourcing

Cloud & Infrastructure: AWS, Docker, Docker Compose, CI/CD, Git, Next.js, React

EXPERIENCE

Jalgaon Fruit Sales Cooperative Ltd – Agritech Supply Chain Platform

May 2023 – May 2024

Software Engineer

Jalgaon, India

- Decomposed a legacy PHP, SQL monolith by building Python Flask microservices backed by PostgreSQL, orchestrating automated cron jobs to synchronize state and cutting query latency by **40%** across **1,000+ daily truck transactions**.
- Doubled peak harvest write throughput (**2x**) and scaled revenue from **\$100K to \$200K** by architecting an isolated write path that eliminated 10–15 minute freezes from MyISAM table locking.
- Reduced support chatbot inference costs by **35%** across inventory lookups by implementing Redis-based semantic prompt caching, model fallback routing, and token compression over warehouse metadata.
- Cut deployment cycle time by **30%** and eliminated production regressions by containerizing microservices with Docker, integrating automated pytest validation suites, and configuring multi-stage CI/CD pipelines.
- Built a web procurement platform backed by RESTful APIs, migrating **150+ daily clerks** across 5 warehouses from manual logs to live stakeholder analytics dashboards.

USC Summer Programs

May 2025 – Jul 2025

Teaching Assistant

Los Angeles, CA

- Led program of **150 students** through Python, Git, AI, and REST API development on robotics hardware, guiding functional prototype delivery on schedule.

SELECTED PROJECTS & OPEN SOURCE

MockPad – Real-Time Collaborative Engine & Multimodal AI Replay

May 2026

Node.js, React, WebSockets, Yjs CRDTs, OpenAI Whisper, MongoDB, LevelDB, Python

[\[Live\]](#) [\[GitHub\]](#)

- Architected a real-time collaborative editor on Yjs CRDTs over custom WebSockets, sustaining **sub-100ms sync latency** for **30+ active users** deployed on a single \$6/month DigitalOcean droplet.
- Cut server peak memory consumption by **80%** with **zero session data loss** by implementing an LRU disk-eviction strategy to LevelDB for inactive collaborative rooms.

CloudBridge – AI Cloud Architecture Advisor & Topology Engine

Apr 2026

Next.js, TypeScript, Python, FastAPI, React Flow, AWS (Lambda, S3, RDS, DynamoDB)

[\[GitHub\]](#)

- Engineered an LLM-driven synthesis engine in FastAPI mapping natural language constraints into validated AWS topologies with IAM guardrails and automated Terraform export against the **Well-Architected Framework**.
- Built a reactive DAG topology generator in React Flow and TypeScript, delivering **sub-100ms** real-time parameter mutations, cost projections, and multi-service dependency recalculations.

DevStack – Open Source Developer Infrastructure Platform

Mar 2026

TypeScript, Next.js, Node.js, PostgreSQL, Redis, Docker, GraphQL, WebSockets

[\[GitHub\]](#)

- Engineered a durable workflow execution engine on an event-sourced PostgreSQL schema with TypeScript deterministic replay, restoring crash state **without re-triggering side effects**.
- Built a distributed feature-flag targeting engine delivering **sub-5ms evaluation latency** via Redis caching and real-time WebSockets, secured with RS256 JWT auth and a federated GraphQL gateway.

HONORS & PUBLICATIONS

Aura – visionOS Accessibility Assistant. Aug 2025. 2nd place of 24-hour hackathon; built with Swift, SwiftUI, and RealityKit, delivering fully on-device live captions and environmental sound awareness. [\[GitHub\]](#)

Research Publications. *Texture Feature Analysis Using GLCM Matrix (10 citations)*; medical-transcript NLP; LSB image steganography. [\[GitHub\]](#)